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RESEARCH ARTICLE

Silicon-modified biochar suppresses bacterial wilt and enhances rhizosphere soil health via microbiome modulation and carbon sequestration

Sheng Wang · Tian-tian Zhang · Israt Jahan · Bogui Pan · Yixia Cai · Kunzheng Cai

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Abstract

Background and aims Silicon (Si)-modified biochar (MSC) is an innovative soil amendment that combines the benefits of silicon and biochar to enhance soil and plant health. However, the mechanisms behind its impacts on disease suppression, soil carbon sequestration and soil microbiomes remain largely unexplored.

Methods To assess the impact of Si-modified biochar on soil characteristics and microbial dynamics under *Ralstonia solanacearum* stress, a controlled pot experiment was performed. Soil physicochemical properties, carbon fractions, pathogen abundance, and bacterial community composition were evaluated using established biochemical and molecular approaches.

Results Si-modified biochar significantly reduced the abundance of *Ralstonia solanacearum* in soil by 41.3%, and in tomato roots, stems, and leaves by 67.1%, 72.6%, and 65.1%, respectively ($p < 0.05$). The wilt disease index was reduced by 73.4% ($p < 0.01$), indicating effective suppression of soil-borne pathogens. Additionally, biochar application significantly enhanced rhizosphere soil multifunctionality and carbon storage, as evidenced by increases in soil pH, enzyme activities, soil organic carbon (SOC), and readily oxidizable carbon (ROC). Moreover, silicon-modified biochar altered bacterial community structure and increased the modularity of the microbial co-occurrence network. The relative abundances of

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Highlights
- Silicon-modified biochar has stronger effects on controlling bacterial wilt.
- Silicon-modified biochar promoted greater enrichment of beneficial microbes, such as *Bacillus* species.
- More soil organic carbon was sequestrated when silicon-modified biochar was used.
- Silicon-modified biochar improved soil multifunctionality and soil health.

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S. Wang · T. Zhang · I. Jahan · B. Pan · Y. Cai · K. Cai (✉)
College of Natural Resources and Environment, South China Agricultural University, Guangzhou 510642, China
e-mail: kzcai@scau.edu.cn

S. Wang · T. Zhang · I. Jahan · B. Pan · Y. Cai · K. Cai
Key Laboratory of Tropical Agricultural Environment in South China, Ministry of Agriculture and Rural Affairs, South China Agricultural University, Guangzhou 510642, China

S. Wang · T. Zhang · I. Jahan · B. Pan · Y. Cai · K. Cai
Guangdong Provincial Key Laboratory of Eco-Circular Agriculture, South China Agricultural University, Guangzhou 510642, China

B. Pan
College of Life Science and Technology, Jinan University, Guangzhou 510632, China

beneficial taxa such as *Bacillus* and *Xanthomonadaceae* were significantly increased by 482.7% and 52.7% ($p < 0.05$), contributing to enhanced microbial diversity and network stability.
Conclusions Si-modified biochar mitigated bacterial wilt by enriching beneficial microbes, increasing soil enzyme activity, and promoting carbon sequestration, collectively reinforcing rhizosphere soil health.

Keywords Soil-borne diseases · Silicon-modified biochar · Soil carbon storage · Soil microbial diversity · Sustainable agriculture

Introduction

In recent years, the proliferation of soil-borne diseases has increasingly hindered crop production, posing significant risks to sustainable agriculture and global food security (Haas and Defago 2005; Fones et al. 2020; Yan et al. 2022). Recent studies reported that diminished soil health and disruptions in microbial communities were key contributors to soil-borne diseases (Dubey et al. 2019; Li et al. 2022a, b). Bacterial wilt, a highly virulent disease, impacts more than 200 crop species and severely inhibits their growth and yield (Peeters et al. 2013; Jiang et al. 2017). This pathogen (*R. solanacearum*) can penetrate plants through their roots, leading to plant death, and it can survive in the soil long after the host plants have died (Yadeta and Thomma 2013; Katan 2017). While various cultural, chemical, and biological approaches are currently used to control this disease (Nega 2014), they each have limitations. For instance, the effectiveness of physical treatments such as soil heat disinfection can be limited by local weather and soil conditions (Jiang et al. 2017). Breeding disease-resistant plant cultivars is crucial but is limited by the rapid evolution of pathogens and challenges in market acceptance (Stuthman et al. 2007; Ma et al. 2023; Ajaharuddin et al. 2024). Additionally, while antibacterial agents for plants are under development, they have yet to be scaled for widespread use. Thus, there is a critical need for new, sustainable, and cost-effective solutions to effectively manage soil-borne diseases.

In contemporary agricultural production, biochar—a stable, carbon-rich product derived from biomass pyrolysis—has emerged as a promising strategy for improving crop yield, soil fertility, and disease resistanc (Wang et al. 2020a; Poveda et al. 2021). Its porous structure and high surface area not only enhance soil carbon storage and physicochemical properties but also reshape microbial communities by enriching beneficial taxa, thereby improving plant–soil interactions (Manasa et al. 2024). Numerous studies have shown that biochar can suppress soilborne diseases such as bacterial wilt and root rot, yet its efficacy remains variable due to differences in feedstock and pyrolysis conditions (Jaiswal et al. 2019; Tian et al. 2021). Notably, converting invasive species like Canadian goldenrod (*Solidago canadensis* L.) into biochar provides a dual benefit of biomass utilization and pathogen suppression (Feng et al. 2021; Wang et al. 2024a). Beyond these agronomic benefits, biochar's disease-suppressive effects involve multiple mechanisms: its surface properties enable pathogen adsorption and detoxification; it fosters competitive exclusion by promoting antagonistic microbes such as *Bacillus,* which produce antibiotics and induce systemic resistance in plants (Wang et al. 2024a, b; Wang et al. 2025). Silicon modification further enhances these functions by increasing soil pH to levels unfavorable for acidophilic pathogens like *Ralstonia solanacearum*, improving nutrient availability, and strengthening plant tissues via silica deposition (Wang et al. 2025). Additionally, Si-modified biochar fosters resilient microbial networks and stabilizes organic carbon, supporting sustained soil health and crop protection (Wang et al. 2025).

Recent studies have demonstrated that modified or activated biochar can be prepared via various physical, chemical, and microbial methods to enhance its stability and performance (Wang et al. 2015, 2025; Hafeez et al. 2022). Among these, silicon-modified biochar stands out, as silicon is a crucial inorganic component of biochar. This modification not only gains structural stability but also enhances its functionality in promoting plant health and resistance to pathogens (Wang et al. 2024a, b; Wang et al. 2025). Silicon increases the bioavailability of essential nutrients and strengthens plant cell walls, making them less susceptible to pathogen attacks (Xiao et al. 2014; Wang et al. 2019). Studies have highlighted the prepared by incorporating silicon-rich crop residues such as rice husks, not only utilizes agricultural waste but also significantly improves the environmental functions of biochar (Zia et al. 2017; Rizwan et al. 2019; Wang et al. 2019, 2021; Bukhari et al. 2021). The introduction of silicon into biochar increases the

available silicon content in the soil, thereby aiding plant growth and health, and enhancing soil quality and its water retention capacity (Wang et al. 2019, 2021). Aditionally, silicon-modified biochar has shown effectiveness in alleviating metal toxicity and other stresses such as drought, salinity, and extreme temperatures (Islam et al. 2020; Sathe et al. 2021; Ratnadass et al. 2023; Wang et al. 2025). While silicon-modified biochar offers significant advantages in improving soil structure and plant health, and in managing heavy metal pollution, research on its impact on plant diseases and its potential for disease control is remain limited.

This study examines the role and mechanisms of silicon-modified biochar in controlling tomato bacterial wilt. We hypothesized that silicon-modified biochar is more effective in inhibiting tomato bacterial wilt than unmodified biochar, which is attibuted to several mechanisms through reducing pathogen abundance, enhancing soil carbon sequestration, improving soil multifunctionality and shaping soil microbial community.

Materials and methods

Planting material

Invasive plant material from *Solidago canadensis* L. was collected from Mopanzhou, Changsha, China. Sodium silicate ($Na_2SiO_3 \cdot 5H_2O$) was sourced from Guangzhou Chemical Reagent Factory. The *R. solanacearum* strains (Tim17) were provided by Professor Gaoqing Yuan from Guangxi University in Nanning, China. Tomato seeds (Jinfeng No. 1) were acquired from Guangzhou Qiannong Agricultural Science and Technology Development Co., Ltd. Sandy loam (Eutric Cambisol) soil used in the study was collected from the field area of Enping City, Guangdong, China. The basic soil chemical properties are as follows: pH 4.5, soil organic matter (SOM) 16.2 $g \cdot kg^{-1}$, available nitrogen (AN) 50 $mg \cdot kg^{-1}$, available phosphorous (AP) 26 $mg \cdot kg^{-1}$, available potassium (AK) 73 $mg \cdot kg^{-1}$, total nitrogen (TN) 0.9 $g \cdot kg^{-1}$, total phosphorous (TP) 1.1 $g \cdot kg^{-1}$, and total potassium (TK) 21.5 $g \cdot kg^{-1}$.

Preparation of the Si-modified biochar

The biomass of *Solidago canadensis* L. was collected and used for biochar preparation. The feedstock was rinsed with deionized water and dried at 80℃ for 48 h. The dried biomass was ground to pass through a 1 mm sieve in a cutting mill. To produce unmodified biochar (SC), the ground material was pyrolyzed in a nitrogen atmosphere at 450℃for 2 h in a muffle furnace. Si-modified biochars were synthesized using a soaking method. Initially, 10 g of $Na_2SiO_3 \cdot 5H_2O$ was dissolved in 100 mL of ultrapure water; subsequently, *S. canadensis* L. straw powder was dispersed in the solution and stirred continuously for 3 h at 150 rpm. The mixture was then filtered through a vacuum filter pump to obtain solids, subsequently dried at 80℃ for 48 h. The powdered biomass underwent pyrolysis with nitrogen in a muffle furnace at 450℃ for 2 h to produce Si-modified biochar (MSC). After pyrolysis, both unmodified and Si-modified biochars were further ground and passed through a 100-mesh sieve to ensure uniform particle size for subsequent experiments. The basic chemical properties of the unmodified and Si-modified biochars are listed in Table 1.

Growing conditions

Tomato seeds were surface sterilized by immersing them in a 10% H_2O_2 solution for 10 min, followed by three rinses with ultrapure water. The sterilized seeds were then germinated on petri dishes with dual filter paper layers, maintaining a temperature of 30℃ for a period of 48 h. Seeds were carefully transferred to sterilized peat soil in plug trays sourced from Klasmann, Germany. The incubation process took place in an MGC-400B incubator at a temperature of 30℃, with a relative humidity of 80% and a day-night cycle of 12 h.

Table 1 Basic chemical characteristics of unmodified and Si-modified biochar

	C (%)	H (%)	O (%)	Si (%)	pH	CEC ($cmol^+ \cdot kg^{-1}$)	Ash content (%)	BET ($m^2 \cdot g^{-1}$)
SC	73.48	3.33	14.25	3.48	10.89	28.63	19.93	4.37
MSC	42.94	3.36	25.65	27.31	10.18	21.27	22.40	184.95

SC unmodified biochar, *MSC* Si-modified biochar

The seedlings were watered daily using a half-strength tomato-specific nutrient solution, consisting of 5 mM $Ca(NO_3)_2$, 1.88 mM K_2SO_4, and other essential minerals. Once the seedlings reached the third-leaf stage, they were transplanted into 16.5 cm×17.0 cm pots, with each pot accommodating two seedlings in 2 kg of sterilized peat soil.

Experimental design

The experiment was conducted using a completely randomized design, comprising five treatments with four replicates each. Each replicate consisted of 30 tomato plants. The treatments were: (1) healthy control (CK); (2) *R. solanacearum* inoculation (CR); (3) 10% sodium silicate solution and *R. solanacearum* inoculation (SN); (4) 1% unmodified biochar and *R. solanacearum* inoculation (SC); (5) 1% Si-modified biochar and *R. solanacearum* inoculation (MSC). The 1% (w/w) biochar application rate was selected based on our previous studies, which demonstrated it to be the most effective concentration for promoting plant growth and suppressing disease (Wang et al. 2024a, b; Wang et al. 2025).

Tomato plants were inoculated at the sixth-leaf stage using *R. solanacearum* pathogen strains (Tim 17). These strains were meticulously cultivated on LB medium, which consists of casamino acids (1 $g{\cdot}L^{-1}$), difco peptone (10 $g{\cdot}L^{-1}$), and yeast extract (1 $g{\cdot}L^{-1}$). The strain was cultivated at 28℃ with shaking at 150 rpm for 48 h.. After incubation, the bacterial cells were harvested by centrifugation and washed three times with deionized water. The suspension concentration was measured with a Shimadzu UV-2600 spectrophotometer (Kyoto, Japan), operating at a wavelength of 600 nm. An optical density (OD) of 0.06 was recorded, corresponding to approximately 10^8 colony-forming units per milliliter ($cfu{\cdot}mL^{-1}$). To initiate infection, 50 mL of the bacterial suspension was evenly distributed across each pot. The disease index was meticulously recorded from the initial leaf wilting and monitored over 10 days. Additionally, plant and soil samples were collected for analysis after 10 days of pathogen inoculation.

Determination of disease index and the abundance of R. solanacearum

Plant and rhizosphere soil samples were collected 10 days after pathogen inoculation. Rhizosphere soil was obtained by gently shaking off loosely attached soil particles from the roots. The disease index of bacterial wilt was calculated (Chen et al. 2015). The abundance of *R. solanacearum* in plant tissues and soil was quantitatively determined using quantitative real-time PCR with specific primers. Detailed information and data analysis are presented in the Supplementary Information Methods I.

Analyses of soil properties and multifunctionality in the rhizosphere

Soil moisture content was determined gravimetrically by oven-drying fresh soil at 105 ℃ for 24 h. Soil temperature was measured in situ using a handheld soil thermometer. Soil pH was measured in a 1:2.5 soil-to-water suspension using a calibrated pH meter. Electrical conductivity (EC) was measured in a 1:5 soil-to-water extract using a conductivity meter. Cation exchange capacity (CEC) was determined using the ammonium acetate (1 $mol{\cdot}L^{-1}$, pH 7.0) saturation method. Available phosphorus (AP) was measured by the molybdenum blue colorimetric method after extraction with 0.5 $mol{\cdot}L^{-1}$ $NaHCO_3$. Available potassium (AK) was extracted with 1 $mol{\cdot}L^{-1}$ NH_4OAc and measured using flame photometry. Alkali-hydrolyzable nitrogen (AN) was determined using the alkali hydrolysis diffusion method. All procedures followed the standardized protocols described by Bao et al. (2006).

Soil catalase and sucrase activity were measured using potassium permanganate titration method and 3,5-dinitrosalicylic acid colorimetric method, respectively (Jorge-Mardomingo et al. 2013; Zhao et al. 2016). The p-nitrophenol colorimetric techniques were employed to measure soil β-glucosidase and chitinase activities (Cui et al. 2013). The method for determining acid phosphatase activity was evaluated using the phenol release method (Eivazi and Tabatabai., 1977).

The soil multifunctionality was calculated according to the reference (Qiu et al., 2021), with the soil EC, CEC, AN, AK, AP, SOC, available Si as metrics. Prior to analysis, the Shapiro–Wilk test was applied to assess normality, and non-normal data were transformed using logarithmic or square root transformations. Data not conforming to normality were transformed using logarithmic or square root transformations to approximate a normal distribution.

Whenever a variable contained negative values, the minimum value from the entire dataset was subtracted to obtain a positive version. All variables were standardized to a scale from 0 to 1, and the mean transformed values were calculated to determine the multifunctionality value.

Soil carbon bank index calculation

Soil microbial biomass carbon (MBC) was determined using the chloroform fumigation method according to Lu et al. (2000); soil particulate organic carbon and mineral-associated organic carbon components were separated and measured according to the methods described by Cambardella et al. (1992). Labile carbon was determined by oxidation with a 0.333 M potassium permanganate solution (Blair et al. 1995), while soil organic carbon (SOC) and water-soluble organic carbon (SWOC) were measured using the potassium dichromate external heating method (Bao 2006). The methods for measuring light and heavy fractions of organic carbon in soil were based on Janzen et al. (1992). In this study, experimental soil was used as the reference soil (Blair et al. 1995), the content of soil organic carbon was 16.2 $g \cdot kg^{-1}$. The calculations were performed as follows in the Supplementary Information Method II.

Soil microbial community structure and diversity

Total microbial genomic DNA was extracted from 20 samples using the E.Z.N.A.® soil DNA Kit (Omega Bio-tek, Norcross, GA, U.S.) according to the manufacturer's instructions. Detailed information and data analysis are presented in the Supplementary Information Methods III.

Statistical analysis

All data were analyzed using SPSS 26.0 (SPSS Inc., Chicago, IL, USA). Data normality and homogeneity of variance were tested and visualized using GraphPad Prism 9.0. Pearson correlation analysis was conducted to examine associations among variables. For path analysis, original data were standardized and subjected to dimensionality reduction using IBM SPSS Amos 26.0. Co-occurrence network analysis was performed based on Spearman correlation matrices using the "psych" and "igraph" packages in R (version 4.3.1). Only correlations with $|r| > 0.6$ and $P < 0.05$ were retained to construct the network. The network topology parameters, including degree, closeness, and betweenness, were computed using the "igraph" package. The modularity index was calculated following the Louvain algorithm to identify community structures within the microbial network. Figure visualizes the microbial co-occurrence patterns under different treatments, highlighting changes in network complexity and modularity.

Results

Plant disease index and *R. solanacearum* abundance

Both the plant disease index and the abundance of *R. solanacearum* were all reduced significantly when treated with silicon (SN), unmodified biochar (SC), and Si-modified biochar (MSC), and the most pronounced effects were obtained by Si-modified biochar (Fig. 1). Compared with the treatment of CR, treatments of SN, SC, and MSC significantly reduced the plant disease index by 36.5%, 39.2%, and 80.2%, respectively, at 5 days post-infection (dpi), and MSC reduced it by 73.4% at 9 dpi (Fig. 1A). Furthermore, compared to CR, MSC significantly decreased the abundance of *R. solanacearum* in roots, stems, and leaves by 67.1%, 72.6%, and 65.1%, respectively (Fig. 1B), and it also markedly decreased that of *R. solanacearum* in rhizosphere soil by 41.3% (Fig. 1C). Notably, there is a positive correlation between the disease index and the abundance of *R. solanacearum* in plant tissues and soil (Fig. S1).

Soil organic carbon components, distribution ratios, and soil carbon pool index

Figure 2 showed that SC and MSC had significant effects on soil organic carbon components. Compared to CR, SN increased the contents of SWOC and MOC by 91.2% and 28.1%, respectively. Additionally, compared to SC, MSC significantly increased MBC, SWOC, ROC, MOC, and heavy POC by 85.5%, 489.3%, 14.5%, 14.0%, and 22.1%, respectively. Meanwhile, it significantly reduced SOC, POC, and NLC by 24.2%, 58.7%, and 23.2%, respectively.

In contrast to CK, CR and SN had no significant effects on the indices of CA, AI, CPI, and CPMI.

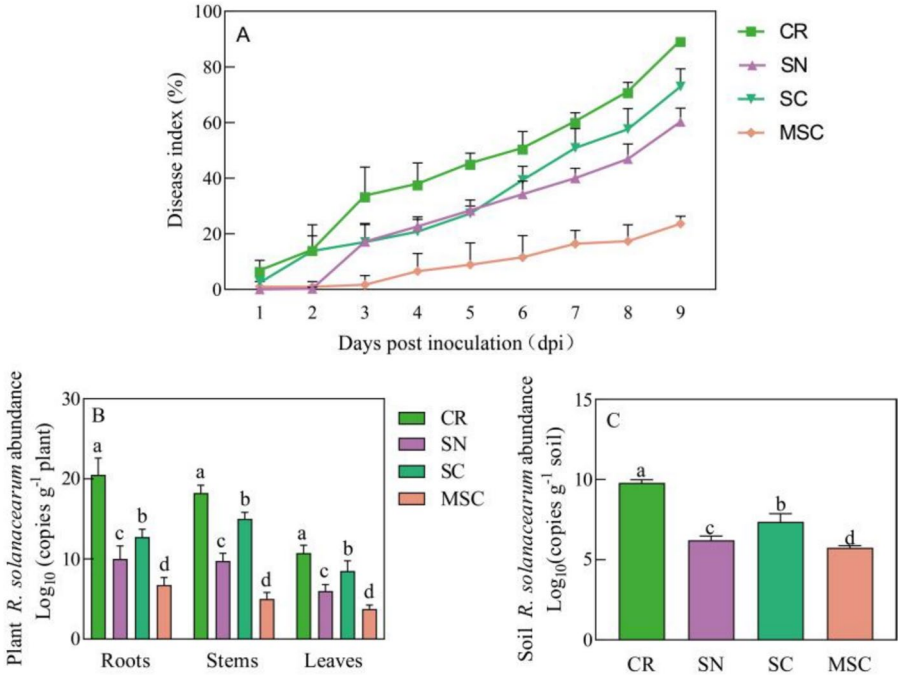

Fig. 1 Effects of Si-modified biochar on plant disease index and abundance of *R. solanacearum*. **A**, Plant disease index; **B**, *R. solanacearum* abundance in plant tissues; **C**, *R. solanacearum* abundance in soil. CR, *R. solanacearum* inoculation; SN, 10% sodium silicate solution and *R. solanacearum* inoculation; SC, 1% unmodified biochar and *R. solanacearum* inoculation; MSC, 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$). Different letters in the same column indicate significant differences at $P < 0.05$

However, MSC significantly increased CPI and CPMI (Fig. 3). Additionally, compared with SC, MSC significantly increased the indices of CA, AI, and CPMI by 62.4%, 62.4%, and 23.2% respectively, while reduced CPI by 24.2%.

In terms of the distribution ratios of organic carbon (Tabel S1), compared with CR, MSC significantly increased the ratio of MBC/SOC, SWOC/SOC, and MOC/SOC ratio, while significantly decreased the POC/SOC ratio. The SC treatment significantly increased NLC/SOC, but reduced ROC/SOC, MOC/SOC, light POC/SOC, and heavy POC/SOC. Notably, compared with SC, MSC significantly increased MBC/SOC, SWOC/SOC, ROC/SOC, MOC/SOC, light POC/SOC, and heavy POC/SOC by 144.6%, 673.5%, 50.4%, 49.7%, 30.7%, and 61.0%, respectively, while reduced POC/SOC and NLC/SOC by 46.3% and 6.9%, respectively.

Soil physical and chemical properties, enzyme activity, and multifunctionality

Si-modified biochar significantly improved soil physicochemical properties and soil fertility (Table 2). Compared to SC, MSC significantly increased soil temperature, moisture, pH, EC, and CEC by 11.8%, 105.3%, 25.0%, 375.0%, and 115.1%, respectively. In addition, soil pH of SN, SC, and MSC were all increased significantly when compared with CR.

In comparison to CK, CR significantly decreased the activities of soil acid phosphatase and chitinase, while increased that ofsoil catalase (Fig. 4). However, compare to CR, the treatments of SN, SC, and MSC significantly enhanceed soil acid phosphatase (96.0%, 85.5%, and 76.5%), catalase (18.3%, 13.0%, and 43.5%), and sucrase activities (86.9%, 215.7%, and 283.9%). In addition, both unmodified and silicon-modified biochar significantly improved soil multifunctionality by 15.4% and 53.4%, respectively. Notably, MSC was superior to other treatments in enhancing chitinase, catalase, and sucrase activities, as well as in improving soil multifunctionality.

Soil microbial diversity, composition and structure

Soil microbial community diversity, composition, and structure were affected significantly by Si-modified biochar (Fig. S3). Shannon diversity indices, Chao diversity indices, and abundance-based coverage estimator metric (ACE) richness estimators treated by CR, SN and SC were much higher than that by CK (Figs. S2A, 2C, and 2D). While the MSC treatments had highest Simpson diversity indices among all treatments (Fig. S2B).

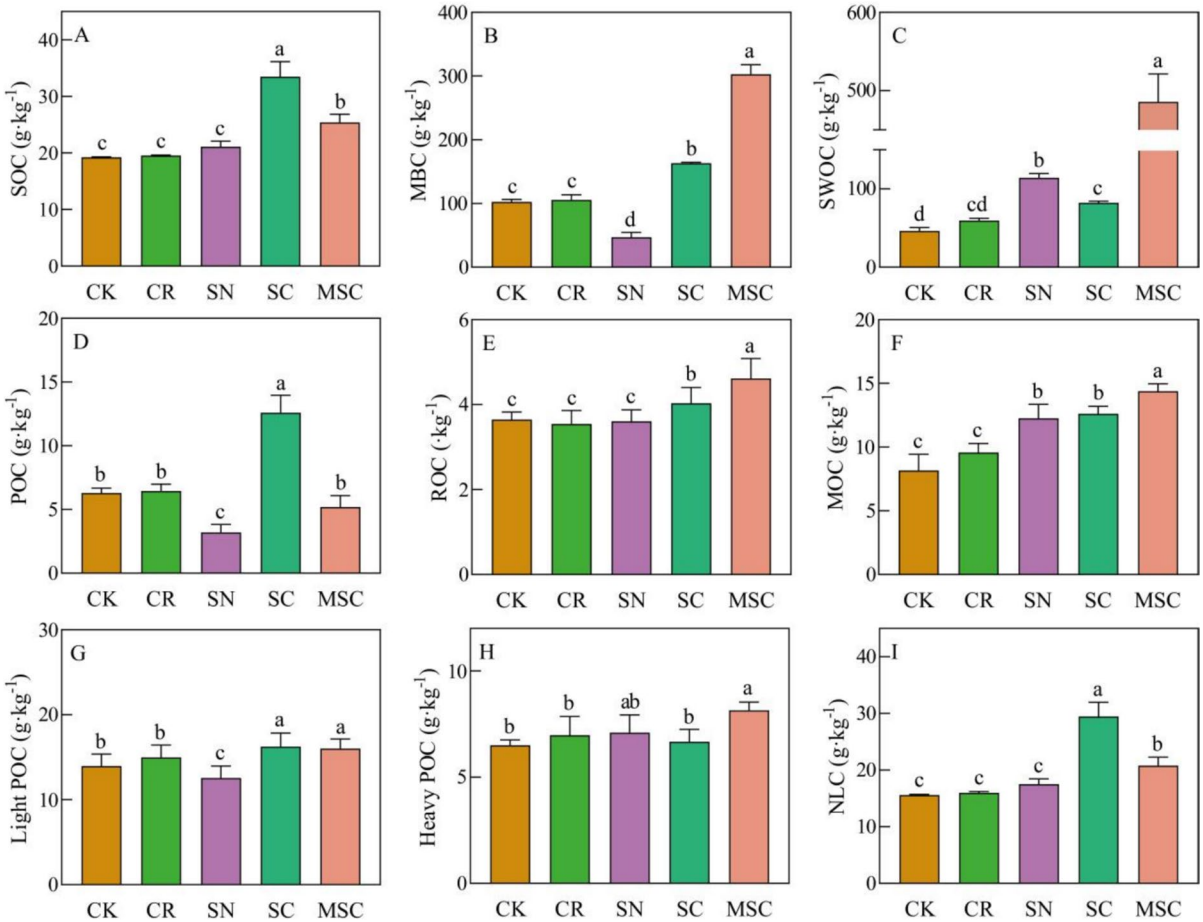

Fig. 2 Effects of Si-modified biochar on soil organic carbon components. **A**, SOC; **B**, MBC; **C**, SWOC; **D**, POC; **E** ROC; **F**, MOC; **G**, Light POC; **H**, Heavy POC and **I**, NLC. CK, healthy control; CR, *R. solanacearum* inoculation; SN, 10% sodium silicate solution and *R. solanacearum* inoculation; SC, 1% unmodified biochar and *R. solanacearum* inoculation; MSC, 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$). Different letters in the same column indicates significant differences at $P < 0.05$

Soil bacterial community components were analyzed at phyla and genus level (Fig. 5A). Silicon (SN) and unmodified biochar (SC) significantly increased the abundance of Actinobacteriota (16.8%, 25.3%) and Gemmatimonadota (48.8%,55.4%) under *R. solanacearum* inoculation. Compared with SC, MSC treatment significantly reduced the abundance of Firmicutes (238.0%), Bacteroidota (206.7%), and Gemmatimonadota (141.5%), but increased the relative abundancy of *Bacillus* (482.7%) and *Xanthobacteraceae* (52.7%) (Fig. 5B). Principal component analysis revealed significant differences in bacterial communities between Si-modified and unmodified biochar at phylum and genus levels, indicating a significant impact of Si modification on soil bacterial communities (Fig. 5C-D).

Co-occurrence network and potential functional prediction of soil bacteria community

Compared to CK, CR increased the number of edges and nodes of soil bacterial co-occurrence network by 5.7% and 6.9%, respectively, while reduced the network densit by 4.0%. Compared to CR, SN, SC, and MSC reduced the nodes of the bacterial co-occurrence network, but increased the density of the bacterial co-occurrence network. Additionally, compared to SC, MSC significantly reduced the number

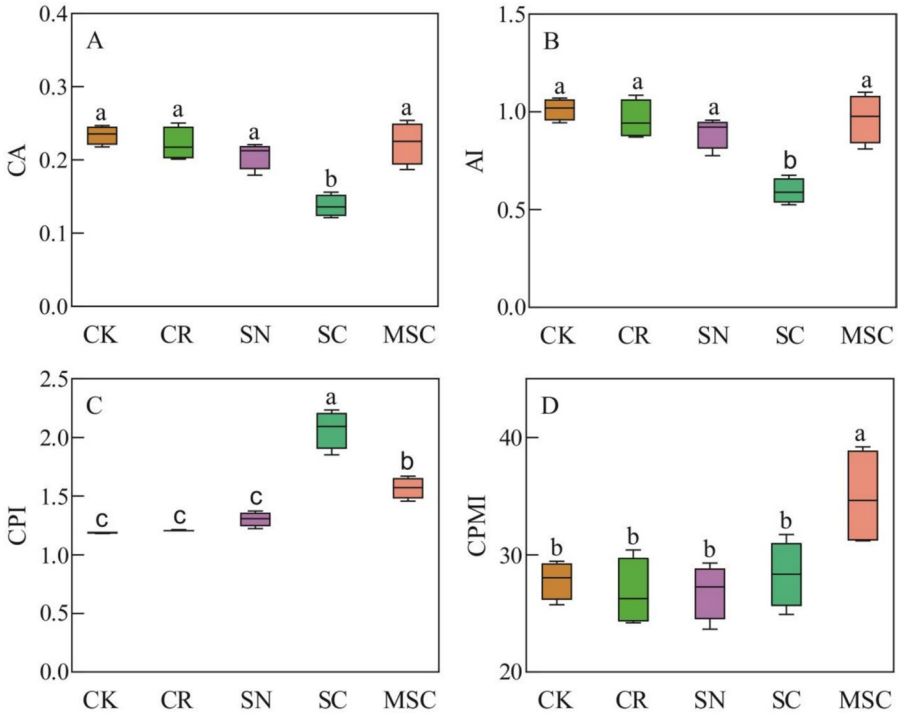

Fig. 3 Effects of Si-modified biochar on the soil carbon pool index. **A**, CA(Carbon activity index); **B**, AI (Activity index); **C**, CPI(Carbon pool index); **D**, CPMI (Carbon pool management index). CK, healthy control; CR, *R. solanacearum* inoculation; SN, 10% sodium silicate solution and *R. solanacearum* inoculation; SC, 1% unmodified biochar and *R. solanacearum* inoculation; MSC, 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$). Different letters in the same column indicates significant differences at $P < 0.05$

Table 2 Effects of Si-modified biochar on the chemical characteristics of the rhizosphere soil

Treatment	Temperature (℃)	Moisture (%)	pH	Electrical conductivity (mS cm^{-1})	CEC ($cmol^{+}\cdot kg^{-1}$)	Available N ($mg\cdot kg^{-1}$)	Available P ($mg\cdot kg^{-1}$)	Available K ($mg\cdot kg^{-1}$)
CK	34.3 ± 0.5bc	0.16 ± 0.03d	4.7 ± 0.1d	0.19 ± 0.02bc	4.0 ± 0.3b	48.0 ± 4.5a	24.7 ± 0.3ab	73.7 ± 1.7b
CR	34.8 ± 0.5b	0.31 ± 0.06b	4.6 ± 0.1d	0.23 ± 0.04b	4.2 ± 0.3b	47.2 ± 4.8a	25.6 ± 0.5a	76.0 ± 2.2b
SN	34.5 ± 0.6bc	0.26 ± 0.04bc	5.5 ± 0.0b	0.16 ± 0.01 cd	4.6 ± 0.4b	48.0 ± 4.5a	22.4 ± 1.4c	76.0 ± 1.8b
SC	34.0 ± 0.0c	0.19 ± 0.04 cd	5.2 ± 0.0c	0.12 ± 0.02d	5.3 ± 0.4b	50.3 ± 6.2a	25.1 ± 1.3ab	113.0 ± 4.2a
MSC	38.0 ± 0.0a	0.39 ± 0.06a	6.5 ± 0.2a	0.57 ± 0.05a	11.4 ± 1.9a	49.0 ± 5.0a	23.0 ± 1.7bc	109.8 ± 2.6a

CK, healthy control; CR, *R. solanacearum* inoculation; SN, 10% sodium silicate solution and *R. solanacearum* inoculation; SC, 1% unmodified biochar and *R. solanacearum* inoculation; MSC, 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$). Different letters in the same column indicates significant differences at $P < 0.05$

of edges and nodes by 17.8% and 15.8%, respectively, but notably increased the network densit by 15.8% Figure 6.

The interrelationship between soil chemical properties, microbial characteristics, and disease factors

In this study, it was found that the abundance of *R. solanacearum* in soil and the plant disease index were significantly negatively correlated with soil pH, CEC, chitinase, sucrase, and peroxidase activity, while positively correlated with soil β-glucosidase activity. Moreover, these disease indicators exhibited significant negative correlations with SWOC, ROC, and MOC. At the phylum level, Proteobacteria, Bacteridota, and Gemmatimonadota exhibited negative correlations with the abundance of *R. solanacearum* in soil and the disease index, while Chloroflexi and WPS-2 exhibited positive correlations. At the genus level, the abundance of soil *R. solanacearum* was negatively correlated with the abudance of *Bacillus* and *Xanthobacteraceae*,

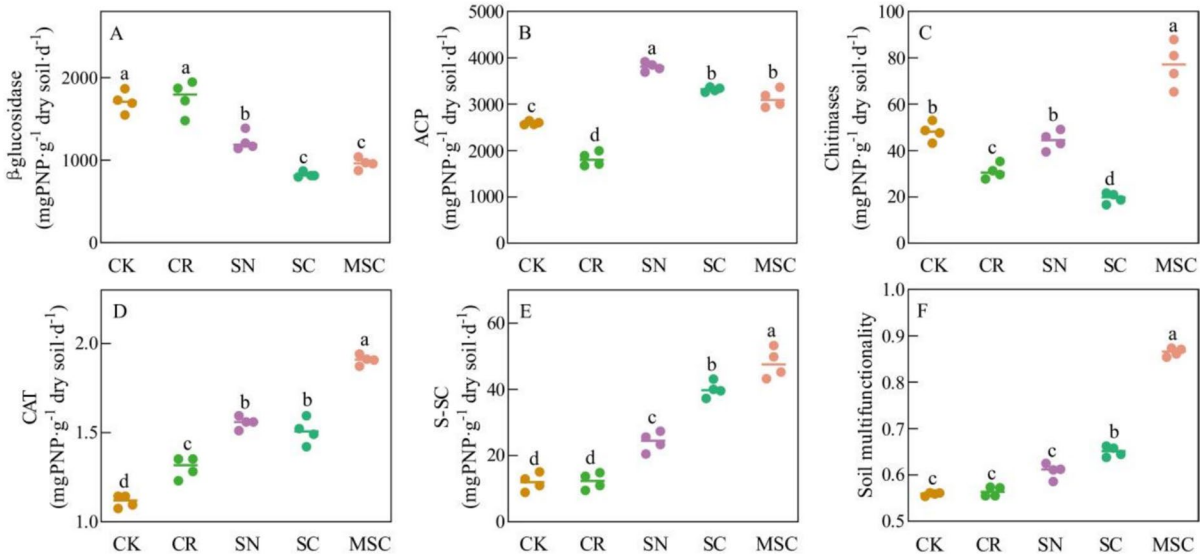

Fig. 4 Effects of Si-modified biochar on the soil β-glucosidase (**A**), acid-phosphatase (**B**), chitinase (**C**), catalase (**D**), sucrase activities (**E**) and multifunctionality (**F**). CK, healthy control; CR, *R. solanacearum* inoculation; SN, 10% sodium silicate solution and *R. solanacearum* inoculation; SC, 1% unmodified biochar and *R. solanacearum* inoculation; MSC, 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n=4$). Different letters in the same column indicates significant differences at $P<0.05$

while positively correlated with the *HBS-OF53. F07, LXW8*, and *JG30.KF.AS9* Figure 7.

The data on soil multifunctionality, the presence of *R. solanacearum* in soil, *Bacillus* species, and Simpson diversity indices collectively explained most of the variations observed in plant *R. solanacearum* levels and the associated disease index (Fig. 8). PLS-PM analyses revealed positive effects of soil multifunctionality on soil *Bacillus* (0.9297; $P<0.001$) and Simpson indices (0.5095; $P<0.05$) were very significant (Fig. 8A). Additionally, there was a positively correlation between soil *Bacillus* and Simpson indices (0.5538; $P<0.05$). Furthermore, significant negative correlations were observed between soil *R. solanacearum* and both plant *R. solanacearum* (0.7632; $P<0.001$) and the disease index (0.4278; $P<0.05$). In addition, soil multifunctionality, *Bacillus*, and Simpson indices negatively influenced the abundance of soil and plant *R. solanacearum* (Fig. 8B).

Discussion

Si-modified biochar strongly enhances the suppression of tomato wilt disease

Si-modified biochar demonstrated the most effective suppression of tomato bacterial wilt (Fig. 1A). This phenomenon could be attributed to the synergistic effects of silicon and biochar on plant and soil health. The pathogen *R. solanacearum* primarily infected plants through the roots, with its abundance in the soil presenting a significant threat due to its ability to infiltrate plant roots and impact the entire plant (Puigvert et al. 2017; Tian et al. 2021). This study showed that the application of Si-modified biochar substantially reduces the abundance of soil *R. solanacearum*, thereby decreasing its colonization in plants. This reduction was a clear indication of enhanced soil health due to the introduction of Si-modified biochar.

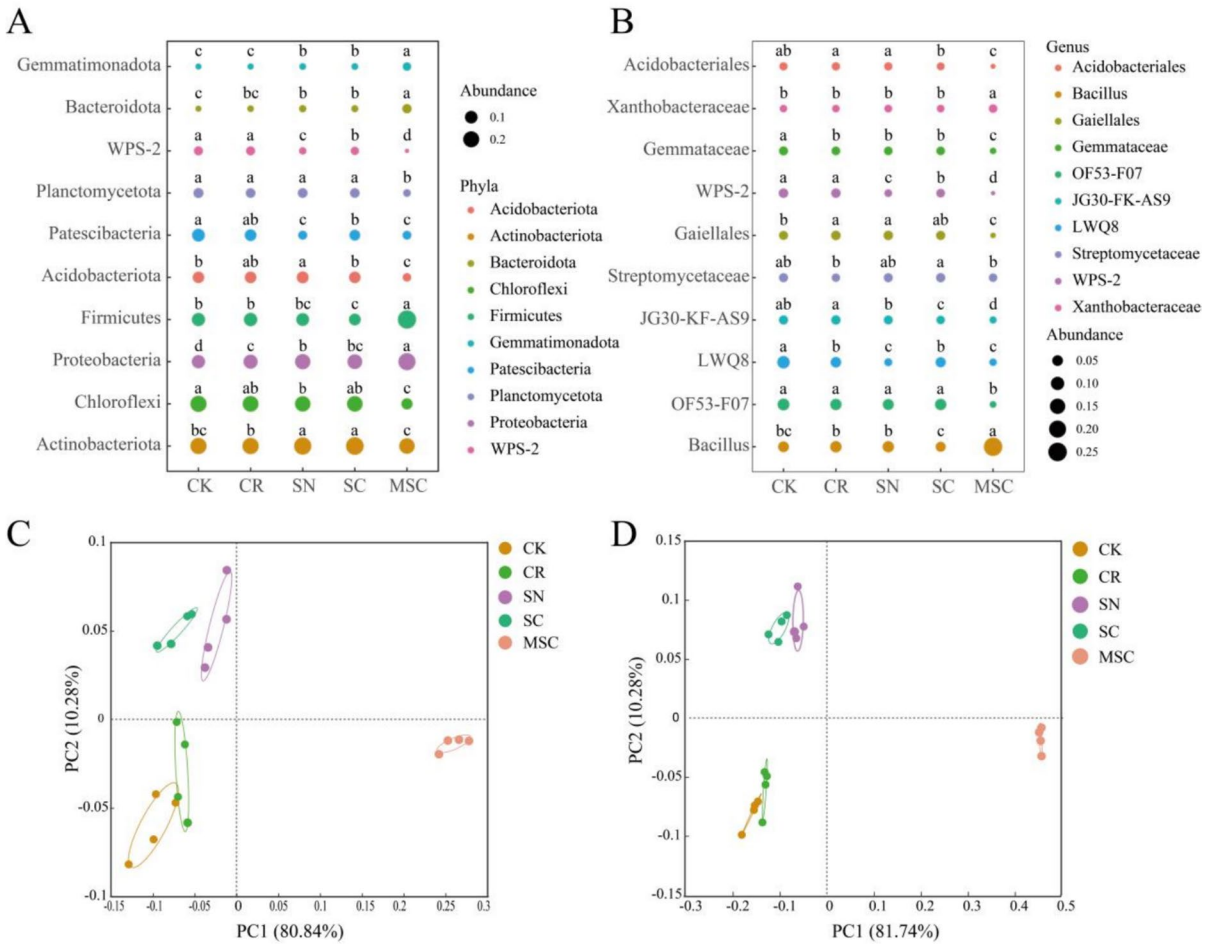

Fig. 5 Effects of Si-modified biochar on abundance of soil microbial taxa at phyla level (**A**) and genus level (**B**), and Principal Component analysis (PC) of soil microbial communities at phyla (**C**) and genus level (**D**). CK healthy control; CR *R. solanacearum* inoculation; SN 10% sodium silicate solution and *R. solanacearum* inoculation; SC 1% unmodified biochar and *R. solanacearum* inoculation; MSC 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$). Different letters in the same column indicate significant differences at $P < 0.05$

Consistent with previous research, biochar had been shown to lower pathogen levels in both soil and crops, thereby mitigating plant diseases (Poveda et al. 2021; Tian et al. 2021). Furthermore, the abundance of *R. solanacearum* in soil and plants showed a positive correlation with the disease index. That is, as the abundance of *R. solanacearum* decreases, the disease index or incidence of bacterial wilt also correspondingly decreases. Zhao et al. (2017) showed that reducing the colonization of pathogenic bacteria in soil is related to lower occurrence of diseases. Elsayed et al. (2020) confirmed that the suppression of bacterial wilt was associated with reduced *R. solanacearum* populations in rhizosphere soil. In summary, Si-modified biochar not only improves soil quality and promotes plant growth but also significantly enhances the resistance to pathogens in soil–plant system. This dual function highlights its potential as an effective strategy for sustainable agriculture by enhancing soil quality and reducing disease incidence in crops.

Silicon-modified biochar promotes soil multifunctionality

Soil health is defined as the capacity of soil to function as a dynamic, biodiverse ecosystem that sustains terrestrial life (Doran et al., 2000). It is commonly

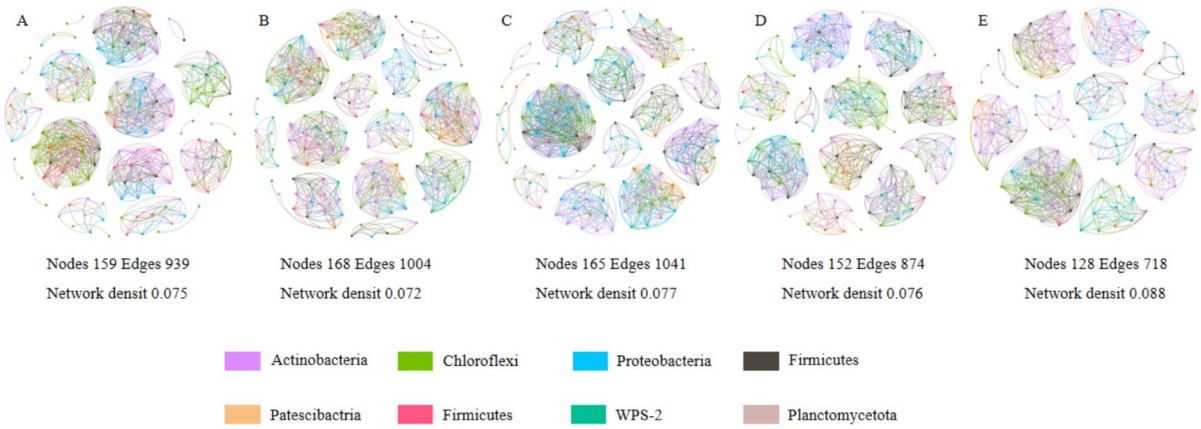

Fig. 6 Effects of Si-modified biochar on co-occurrence network analysis of bacterial communities in rhizosphere soil. **A**, CK; **B**, CR; **C**, SN; **D**, SC; **E**, MSC. CK healthy control; CR *R. solanacearum* inoculation; SN 10% sodium silicate solution and *R. solanacearum* inoculation; SC 1% unmodified biochar and *R. solanacearum* inoculation; MSC 1% Si-modified biochar and *R. solanacearum* inoculation. All values are presented as the mean ± standard error ($n = 4$)

assessed through a suite of physical, chemical, and biological indicators. Maintaining optimal soil health is fundamental to suppressing soil-borne pathogens (Wang et al. 2019).. In this study, Si-modified biochar (MSC) significantly increased soil pH, effectively neutralizing acidity compared to the SN and SC treatments and creating an alkaline environment that is unfavorable for the survival of *Ralstonia solanacearum* (Hou et al. 2022; Hafeez et al. 2022). This alkalizing effect is attributable to the presence of basic mineral components in biochar—such as silicates and hydroxides—which exert liming effects and enhance both cation exchange capacity and nutrient availability. Elevated soil pH directly inhibits acidophilic pathogens, thereby reducing disease risk.

Simultaneously, Si-modified biochar markedly improved soil water retention capacity, likely due to enhanced porosity and hydraulic conductivity (Schaller et al. 2020). Meta-analytical evidence indicates that biochar application significantly increases soil water-holding capacity, field capacity, and total porosity, which in turn supports microbial proliferation and functional activity under drought conditions (Edeh et al. 2020). In parallel, Si-modified biochar substantially boosted soil enzymatic activity. Increases were observed in sucrase and peroxidase, with a particularly pronounced elevation in chitinase—a lignin-degrading enzyme crucial for antifungal defense. Chitinase hydrolyzes fungal cell walls and serves as a key component of microbial-mediated pathogen suppression (Berini et al. 2018). The enhanced enzyme activities are likely driven by organic carbon enrichment, which provides both substrates and energy for microbial metabolism (Masto et al. 2006; Piotrowska-Długosz., 2020; Li et al. 2022a, b). This is consistent with previous reports showing that carbon-rich environments stimulate microbial enzyme activity. Conversely, a decrease in β-glucosidase activity was noted, potentially due to substrate competition and the hydrophobic surface properties of biochar, which can impede enzyme–substrate interactions (Shukla et al. 2023). Collectively, these findings indicate that Si-modified biochar enhances soil biochemical function and enzymatic defenses, thereby strengthening soil resistance against pathogen invasion.

The diversity and composition of soil microbial communities are central to maintaining soil function and ecological stability (Griffiths and Philippot., 2013). Our results show that Si-modified biochar also reshaped the soil microbiome, significantly increasing bacterial diversity and enriching beneficial taxa such as *Bacillus* and *Xanthobacteraceae*. Notably, the abundance of *Bacillus* was significantly elevated in Si-modified biochar soils. Members of this genus are well known for producing antibiotics and lytic enzymes, and for inducing systemic acquired resistance in plants, making them effective suppressors of phytopathogens (Choudhary et al., 2009); Shafi et al. 2017). In addition, many *Xanthobacteraceae* are

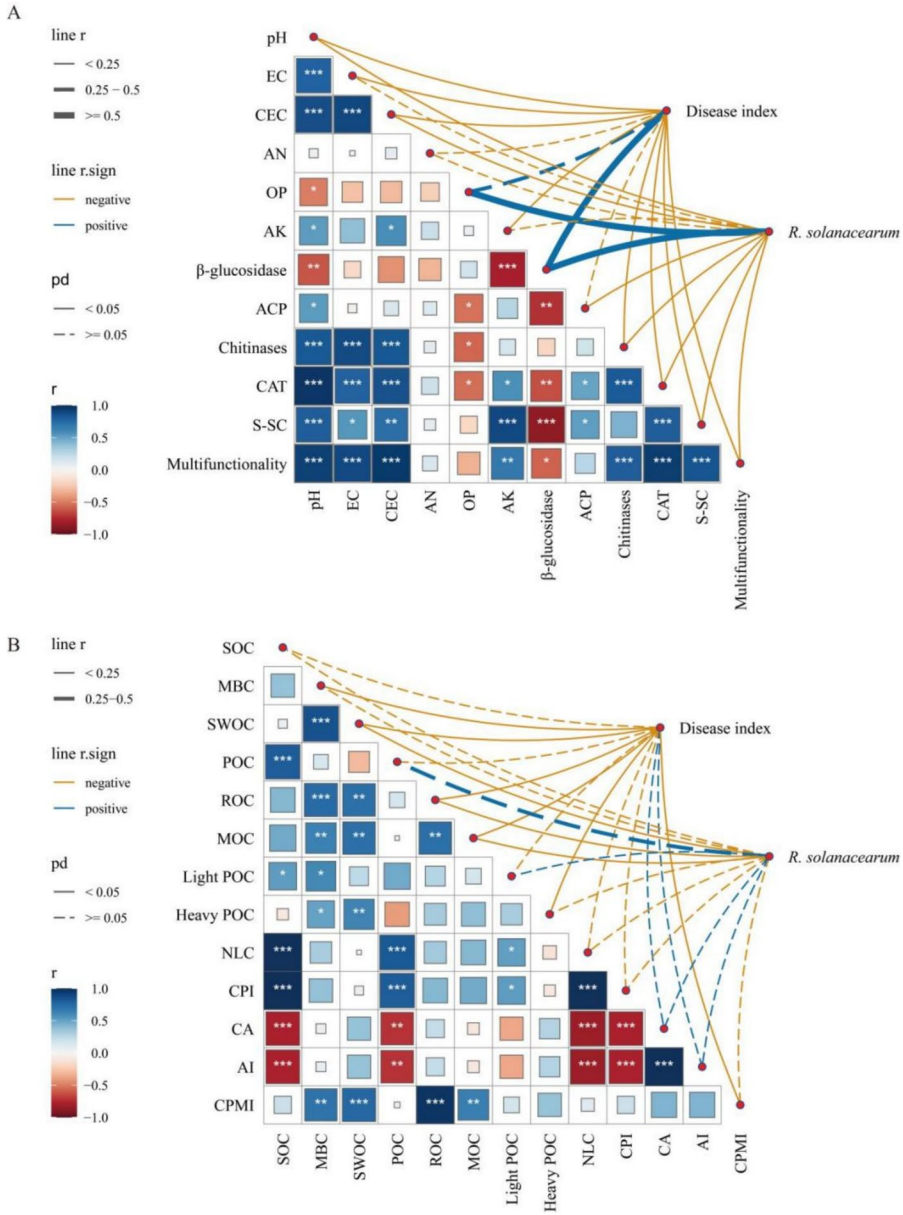

Fig. 7 Correlation analysis between soil chemical characteristics and plant disease index, and soil *R. solanacearum* abundance. **A**, relationship between soil properties and plant disease index as well as *R. solanacearum* abundance; **B**, effects of soil carbon forms, carbon pool and plant disease index as well as R. solanacearum abundance. All values are presented as the mean ± standard error ($n = 4$). The data on the arrows are a direct effect of normalization. * $P < 0.05$, * * $P < 0.01$, and * * * $P < 0.001$

actively involved in nitrogen cycling and the degradation of environmental pollutants (Zhan et al. 2021; Zhou et al. 2023). Importantly, both *Bacillus* and *Xanthobacteraceae* abundances were negatively correlated with *R. solanacearum* abundance, suggesting that Si-modified biochar promotes a disease-suppressive rhizosphere through microbiome restructuring. Co-occurrence network analysis revealed that pathogen infection (CR) increased overall microbial connectivity but led to a sharp decline in network modularity, reflecting an unstable and diffuse emergency response (Ambrosini et al. 2016; Mordecai 2011). In contrast, Si-modified biochar soils exhibited fewer nodes and edges but significantly higher network modularity. According to ecological network theory, high modularity implies the formation of discrete, semi-independent sub-networks that can buffer disturbances locally without propagating them system-wide, thereby enhancing system resilience (Widder et al. 2016; Banerjee et al. 2018; Liu et al. 2020).. Thus, Si-modified biochar fosters a soil microecosystem that is both functionally diverse and structurally

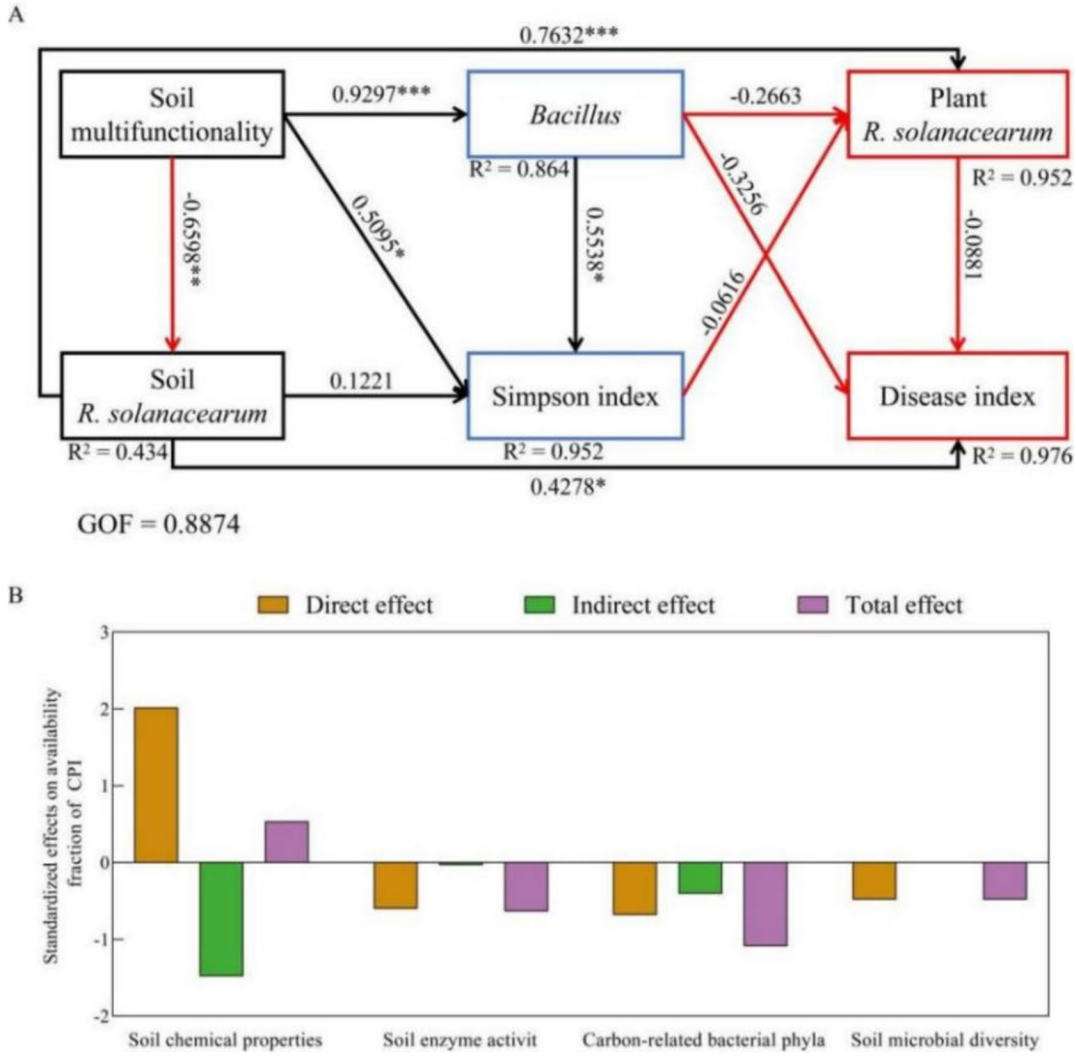

Fig. 8 Partial least squares path analysis shows how soil fertility, *R. solanacearum* abundance, and soil bacterial diversity impact plant disease index under *R. solanacearum* infection. A, Partial least squares path analysis; B, Standardized effecte on availabilability fraction of disease index. Dotted lines show negative effects and active lines show positive effects. The number and asterisk above the line represent normalized path coefficients and significance levels (*, $P<0.05$; **, $P<0.01$; ***, $P<0.001$)

robust, capable of sustaining long-term disease suppression through improved compartmentalization of microbial functions.

Si-modified biochar can enhance soil organic carbon stability

Soil carbon is a key indicator of soil health, influencing its physicochemical and biological properties, and serving as a cornerstone for soil health assessment frameworks (Lal et al. 2016). Biochar application has been shown to significantly increase soil organic carbon (SOC) (Wang et al. 2020c; Poveda et al. 2021; Wang et al. 2024a, b). In this study, Si-modified biochar significantly enhanced microbial biomass carbon (MBC), salt water-extractable organic carbon (SWOC), readily oxidizable carbon (ROC), mineral-associated organic carbon (MOC), and heavy particulate organic carbon (POC) compared to the SC treatment. These increases are largely attributed to

Si-modified biochar capacity to improve soil moisture and thermal regimes, which are critical (Davidson and Janssens., 2006; Fang et al. 2022). Soil temperature and moisture modulate the decomposition and stabilization of organic matter, especially the more labile SOC fractions like MBC (Krull et al. 2003; Ramesh et al. 2019; Wang et al. 2020b). Si-modified biochar porous structure enhances water retention, slowing organic matter decomposition and supplying a sustained carbon source for microbial growth, thereby increasing microbial biomass. Moreover, Si-modified biochar facilitates the physical protection of organic carbon through adsorption and microaggregate formation with soil minerals, enhancing SOC stability. Its high aromatic carbon content further resists microbial oxidation, reducing SOC mineralization (Xiao and Chen., 2017; Riaz et al. 2017; Han et al. 2016; Poveda et al. 2021).

Beyond improving SOC fractions, Si-modified biochar significantly influenced the Carbon Pool Index (CPI) and Carbon Pool Management Index (CPMI). These effects are likely mediated by changes in soil chemical properties and microbial activity, including enhanced proliferation of beneficial microbes and stabilization of organic carbon pools, ultimately improving carbon allocation and utilization. Prior studies have shown that biochar boosts both carbon sequestration and microbial activity (Karimi et al. 2020; Rahman et al. 2020; Poveda et al. 2021). Additionally, silicon-modified biochar raised the ratios of MBC/SOC, SWOC/SOC, and MOC/SOC significantly. The increase of soil microbial biomass carbon and dissolved organic carbon could improve soil microbial environment and increase the availability of soil organic carbon (Liang et al. 2011; Zhang et al. 2021). A significant reduction in the ratio of POC/SOC ratio suggested that silicon-modified biochar could improve soil structure and promote the formation of soil aggregates, which not only contributed to preventign the rapid decomposition of organic carbon but also fostering long-term carbon storage. Improved soil structure also enhanced water holding capacity, aeration, root penetration, and nutrient uptake, thereby promoting plant growth and overall soil ecosystem health (Sharma and Kumar., 2023; Hartmann et al., 2023). Future research should investigate the long-term impact of silicon-modified biochar on soil stability and its combined effect with other sustainable practices to enhance crop health and productivity.

Conclusions

Our results demonstrated that both silicon and biochar, particularly silicon-modified biochar (MSC), significantly reduced Ralstonia solanacearum abundance in soil and plant tissues, as well as the overall disease index. MSC exhibited the highest efficacy in suppressing bacterial wilt and enhancing soil health by increasing soil pH, improving multifunctionality, and stimulating the activities of key defense-related enzymes including chitinase, sucrase, and catalase. Additionally, MSC significantly elevated levels of soil water-soluble organic carbon (SWOC) and readily oxidizable carbon (ROC), while also promoting bacterial diversity and enriching beneficial microbial taxa such as Bacillus and members of the Xanthobacteraceae family. These shifts collectively restructured the rhizosphere microbial community toward a more suppressive and resilient state, underscoring the potential of MSC to enhance the soil–plant interface and support sustainable disease control. Future research should explore the long-term impacts of MSC across diverse soil types, crop systems, and climatic conditions, alongside assessments of economic feasibility and potential environmental risks such as soil alkalization and silicon accumulation, to ensure both efficacy and safety in large-scale applications.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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